Deepak Dhinwa

Mechanical Engineering | Indian Institute of Technology Bombay

B.Tech. | deepak.dhinwa05@gmail.com | +91 9216999404

Professional Experience

Analytics Intern | *Scapia* | *Indian Travel-Fintech startup* [May '26 - July '26]

- Analyzed **user behavior** from user interaction data to identify usage patterns and potential **card-abuse** activity.
- Built **churn prediction models** to identify at-risk customers across different stages and support targeted retention strategies.
- Developed an end-to-end **hotel highlight-to-image matching pipeline** using fine-tuned **SigLIP2**, with automated inference orchestrated via **Airflow** on **Databricks GPU** compute.
- Analyzed existing **dynamic flight discounting** approaches and collaborated with my team to formulate **hypotheses and experiments** for discount allocation and incremental bookings.

Key Projects

SportMind - NBA Intelligent Analytics Platform *[Jan'26 - Feb'26]*

- Ingested 10 seasons of NBA data (∼1,400 players, ∼12,800 games) into a **Neo4j knowledge graph** and **Weaviate vector store**, enabling hybrid BM25 + dense retrieval with cross-encoder reranking.
- Designed a **RAG orchestrator** using LlamaIndex with intent classification, Cypher graph querying, and **auto-rotating LLM fallback** (Groq + Gemini) with LRU caching.
- Trained an **XGBoost win-probability model** with isotonic calibration and **658 EWMA player forecast models** across 7 metrics for 94 players; deployed a **FastAPI + React** application with live chat, graph exploration, and prediction dashboards.

Real-Time Vehicle Speed Detection & License Plate OCR *[Jun'25 - Jul'25]*

- Built a real-time **vehicle speed estimation system** using **YOLOv8** and **SORT tracking**, achieving ∼15ms inference per frame on RTX 3050 GPU with consistent vehicle ID assignment.
- Trained a **custom YOLOv8** number plate detector on ∼222 cropped US license plate images for 50 epochs, achieving ∼**95% mAP50 and** ∼**67% mAP50-95**; fine-tuned **Microsoft TrOCR** for plate OCR.
- Dockerized the **FastAPI pipeline** integrating YOLOv8, TrOCR, and SORT, and deployed it as a scalable microservice on **Azure Container Registry** tested with live IP webcam feeds.

Autonomous Car Simulation (MetaDrive) *[Aug'25 - Nov'25]*

- Designed a **sensor-fusion policy** combining **EfficientNet-B0** visual features with a 19-dim state MLP, reducing MSE by **81.5%** over a camera-only baseline; trained via Behavioural Cloning on ∼90k IDM expert steps.
- Used **Grad-CAM** to identify attention drift toward sky and terrain as the cause of **95% of out-of-road episodes**; warm-started **PPO** from the BC checkpoint, improving average survival **7.5**× (299 → 2256 steps).
- Exported the policy to **ONNX** (19.6 MB) and built a **Streamlit dashboard** for Grad-CAM heatmaps and BC vs. RL training curves; identified reward sparsity as the main barrier to route completion.

Dual-Task Image Reconstruction & Colorization *[Sep'25 - Nov'25]*

Advanced Topics in Deep Learning for Image Analysis | *Guide: [Prof. Biplab Banerjee]*

- Designed a **multi-task image processing architecture** using **DINOv2 ViT-S/14**, with a shared convolutional decoder and task-specific heads for RGB reconstruction and LAB colorization.
- Applied **LoRA-based parameter-efficient fine-tuning**, freezing pretrained encoder weights while adapting transformer layers and reducing trainable parameters by ∼**90%** for memory-constrained training.
- Trained on **COCO 2017** with 224×224 images and **25% patch masking**, using L1 + SSIM reconstruction loss and LAB a*b* colorization loss; identified **multi-task interference and limited training** as key colorization bottlenecks.

Airline Passenger Satisfaction - Data-Centric ML Pipeline *[Oct'25 - Nov'25]*

- Analyzed missingness across **103,904 records** using the MCAR/MAR/MNAR framework and applied regression-based imputation, achieving $\mathbf{R^2}$ **= 0.93** for arrival-delay prediction.
- Eliminated multicollinearity using iterative **VIF analysis** and reduced 14 survey ratings using **PCA**; compared **Mutual Information, RFE, and L1 Logistic Regression** for feature selection.
- Trained **XGBoost** achieving **ROC-AUC = 0.987 and F1 = 0.935**; used **SHAP** to identify cabin comfort as the dominant satisfaction driver and analyse product-level dissatisfaction patterns.

Technical Skills

Languages	Python, C++, C, SQL
Tools	Git, Docker, VS Code, Postman, Azure, Databricks, Weights & Biases, MLflow, TensorBoard
Libraries & Frameworks	NumPy, Pandas, OpenCV, scikit-learn, PyTorch, TensorFlow, XGBoost, Hugging Face, LangChain, LangGraph, Apache Spark, Apache Airflow